Arista Whitefield POP Access Node WFD-ACC-48 Hardware/Environmental Fault Failure SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3490

Description

This Standard Operating Procedure (SOP) addresses a critical hardware/environmental fault failure on the Arista Access Node WFD-ACC-48 located at the Whitefield POP. The failure domain in question revolves around ASIC memory parity errors and line card crashes, precipitated by power supply redundancy loss. The Arista Access Node, being a critical component in the network's access layer, is responsible for aggregating and forwarding traffic between the campus network and the wider internet. The ASIC memory parity errors and line card crashes on this node can lead to a loss of forwarding capability, resulting in packet drops and connectivity issues. The power supply redundancy loss exacerbates this situation, making it essential to promptly identify and rectify the issue to prevent network downtime and data loss. The root cause of this failure domain can be attributed to various factors, including environmental conditions, hardware malfunctions, or configuration issues. Therefore, it is crucial to meticulously follow this SOP to ensure prompt and effective resolution.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield POP to physically access the node.

- Verify the status of the redundant link to ensure it is operational and healthy.

- Check the environmental conditions of the node, including temperature and humidity levels, to rule out any external factors contributing to the failure.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to hardware malfunction	High
Power supply redundancy loss due to faulty power supply unit	Medium
Configuration issue causing ASIC memory parity error	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal switch internal detail | grep ASIC, Expected Output: ASIC Memory Parity Error: Yes, Description: This command checks for ASIC memory parity errors, which can indicate a hardware malfunction or configuration issue.

Command: show system internal switch internal detail | grep Line Card, Expected Output: Line Card Status: Down, Description: This command checks the status of the line card, which can indicate a hardware malfunction or power supply issue.
```

```
Command: show power status, Expected Output: Power Supply Redundancy:
No, Description: This command checks the power supply redundancy
status, which can indicate a power supply issue.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or line card, Command: replace fabric ASIC, Description: Replace the faulty ASIC or line card with a new one to resolve hardware malfunctions.

Step 2: Step: Replace faulty power supply unit, Command: replace power supply unit, Description: Replace the faulty power supply unit with a new one to resolve power supply issues.

Step 3: Step: Reconfigure the node to resolve configuration issues, Command: configure terminal, Description: Reconfigure the node to resolve configuration issues causing ASIC memory parity errors.

Post-Check Verification

```
Command: show system internal switch internal detail | grep ASIC,
Expected Output: ASIC Memory Parity Error: No, Description: This
command verifies that the ASIC memory parity error has been resolved.

Command: show system internal switch internal detail | grep Line Card,
Expected Output: Line Card Status: Up, Description: This command
verifies that the line card is operational.

Command: show power status, Expected Output: Power Supply Redundancy:
Yes, Description: This command verifies that the power supply
redundancy is operational.
```

Roll-back Procedures

- Step: Restore original configuration, Command: configure replace disk0:/config/config.txt, Description: Restore the original configuration to revert any changes made during the resolution.

- Step: Replace original hardware components, Command: replace fabric ASIC, Description: Replace the new hardware components with the original ones to revert any changes made during the resolution.